

Day 7 — *Pudīnā* (Mint)

Module: Auṣadhi · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students can identify *pudīnā* (*Mentha* species) by leaf shape and the characteristic cooling smell, name **menthol** as the principal volatile, distinguish two common varieties (spearmint vs peppermint), and explain that mint is classified as *śīta* (cooling) in Ayurveda — a classification that aligns with its measurable physiological effect.

Materials

- Two sprigs of mint, ideally from two different varieties (spearmint and peppermint are easiest to source)
- A small jar of dried mint leaves
- A few peppermint candies or a small tube of menthol-containing balm — for the smell bridge (do not let students apply balm to their skin in class)
- Ice cube + a small bowl (for the demo on cooling sensation)
- Whiteboard

Vocabulary introduced today

- *pudīnā* (IAST: pudīnā) — mint. Commonly refers to *Mentha spicata* (spearmint) in Indian kitchens.
- **Menthol** — the principal volatile cooling compound. Most concentrated in peppermint (*Mentha × piperita*); also present in smaller amounts in spearmint.
- *śīta* (IAST: śīta; lit. “cool, cooling”) — the cooling potency (*vīrya*) in Ayurveda. Contrasts with *uṣṇa* (heating).

Lesson flow

Warm-up (5 min)

Pass the mint sprigs along the row. Each student takes a leaf, rubs gently, sniffs.

“Three words for the smell.” (Likely: “fresh,” “cool,” “toothpaste-like.”)

Hold an ice cube briefly — note how cold feels on the skin. Now ask: “When you smelled the mint just now, did it feel like the ice or different?” (Different — but related. We’ll explain.)

Core (20 min)

1. **Identify it.**
 - **Mint family (Lamiaceae)** — same family as tulsi (square stems!). Make students check: press the mint stem between thumb and finger. Four flat sides.

- **Leaf shape:** oval-pointed, serrate (jagged) edges. Larger than tulsi leaves. Bright green.
- **Smell:** the unmistakable cooling-fresh aroma. Different varieties smell slightly different.
- **Common varieties in India:**
 - *Mentha spicata* — spearmint. The everyday “pudina” in chutney, raita.
 - *Mentha × piperita* — peppermint. More concentrated menthol; sharper, “tooth-paste” smell.
 - *Mentha arvensis* — corn mint / Japanese mint — grown commercially in India for menthol extraction.

2. The cooling trick — explained. This is the headline.

- Menthol activates a receptor in your skin called **TRPM8** — the same receptor that activates when your skin actually gets cold.
- The receptor doesn’t know whether the signal is from real cold or from a menthol molecule. It sends “cold!” to your brain either way.
- The result: **you feel cool without your temperature actually dropping.** Your skin temperature is unchanged. Your perception of coolness is real.
- This is the same trick that capsaicin (chilli) plays in reverse — capsaicin activates TRPV1, the “heat” receptor, and your brain feels heat even though your skin isn’t hot.

This is one of the cleanest connections between **traditional classification** (*śīta* — cooling) and **modern molecular pharmacology** (TRPM8 activation). The tradition got the *effect* right; modern science explained the *mechanism*.

3. What mint does — well-supported.

- **Cooling sensation** (TRPM8) — definitively established.
- **Digestive support** — peppermint oil capsules have **good clinical evidence** for relieving symptoms of Irritable Bowel Syndrome (IBS) when prescribed by a doctor. Mechanism: relaxation of gut smooth muscle.
- **Breath freshness** — antibacterial activity in the mouth.
- **Headache relief** when applied to the temples (the cooling sensation distracts from pain). Topical use of dilute menthol; not the same as ingesting essential oil.

4. Cautions.

- **Babies and small children** — undiluted menthol-containing products on the chest or face can cause breathing reflexes. Do not give menthol balms to infants.
- **Peppermint oil capsules** — these are *medications*, not snacks. The IBS evidence is for *prescribed* dosing.
- **Acid reflux** — peppermint can relax the lower oesophageal sphincter, making reflux worse for some people.

5. Traditional Ayurvedic frame. *Pudīnā* is *kaṭu* (pungent) in taste but *śīta* (cooling) in *vīrya*. Used in summer drinks (mint nimbu pani), in chutneys (with coriander and lime), and in some preparations for cooling *pitta* imbalances.

Activity (15 min) — Variety taste-id challenge

(Skip if cooking/tasting is not permitted at your school. Replace with a sketch-by-variety exercise.)

- Set up two cups: spearmint chutney (from prepared chutney, very small portion per student) vs a peppermint candy crushed in water (tiny amount).
- Each pair tries to identify which is which by taste alone.
- Record what made them distinguishable — sharpness? Sweetness? Cooling intensity?

Final 5 min — group discussion: - “*Why does mint feel cool when it isn’t actually cold?*” (TRPM8 activation — students should be able to explain this back.) - “*What’s the connection between the Ayurvedic word *śīta* and this molecular trick?*” (The tradition was tracking the effect; the science explains the cause.)

Wrap-up (5 min)

- Restate the safety rule.
- Preview Day 8: “*Tomorrow we put all six herbs together — and look at them through the *doṣa* lens. Heating, cooling, what goes with what.*”

Student-facing content

Pudīnā (*Mentha* species) — mint.

What it is: A Lamiaceae herb (square stems!). Common varieties in India: *Mentha spicata* (spearmint), *Mentha × piperita* (peppermint), *Mentha arvensis* (corn mint).

How to identify: Oval-pointed serrate leaves, bright green; the unmistakable cooling-fresh smell when crushed.

What’s inside: Menthol — the principal volatile.

Why does mint feel cool? - Menthol activates **TRPM8** — the same receptor in your skin that activates when you actually feel cold. - Your brain receives the “cold!” signal even though your skin’s temperature hasn’t changed. - This is the **mechanism** behind the Ayurvedic classification of mint as *śīta* (cooling) — the tradition observed the effect; modern pharmacology explained the cause.

Well-supported modern uses: - Cooling sensation (mechanism: TRPM8 activation). - Peppermint oil for IBS symptoms (prescribed, not self-administered). - Antibacterial action in the mouth (breath freshness).

Cautions: Not for infants on the chest/face. Concentrated essential oil and capsules are medications, not snacks.

Homework

- Find pudina (or any mint) in your home or near it. Sketch the leaf. Identify which variety if you can. Note: where do you encounter mint most often in your week?

Differentiation

- **One tier up:** TRPM8 was identified as the cold receptor in 2002. Look up the David Julius lab work on TRP channels (2021 Nobel Prize in Physiology or Medicine). Be ready to share one finding next class.
- **One tier down:** Focus on identifying mint by smell and remembering “menthol = cool feeling.”

Teacher notes

- The **TRPM8 mechanism** is the single most rewarding teaching moment of this module for a science-leaning student. The bridge from “cooling herb” (tradition) to “TRPM8 activation” (Nobel-winning science) is satisfying and intellectually honest.
- If anyone in your class connects “menthol cooling = chilli burning = same kind of receptor trick,” that’s the inferential leap to highlight publicly. Praise it.
- A few students may bring up vapour rubs / chest balms used in childhood. Acknowledge the use; note the infant caution; don’t sermonise.

Citation(s) used in this lesson

- Menthol activation of TRPM8: established in molecular pharmacology. David Julius and colleagues identified TRPM8 as the cold/menthol receptor (Julius lab, UCSF; 2021 Nobel Prize work). Reference is at the lab level only; specific paper not cited.
- Peppermint oil for IBS: multiple systematic reviews exist. Teachers can search “peppermint oil IBS systematic review” in PubMed.
- *Pudīnā* as *śīta* — standard Ayurvedic nighantu reference; chapter-level only.